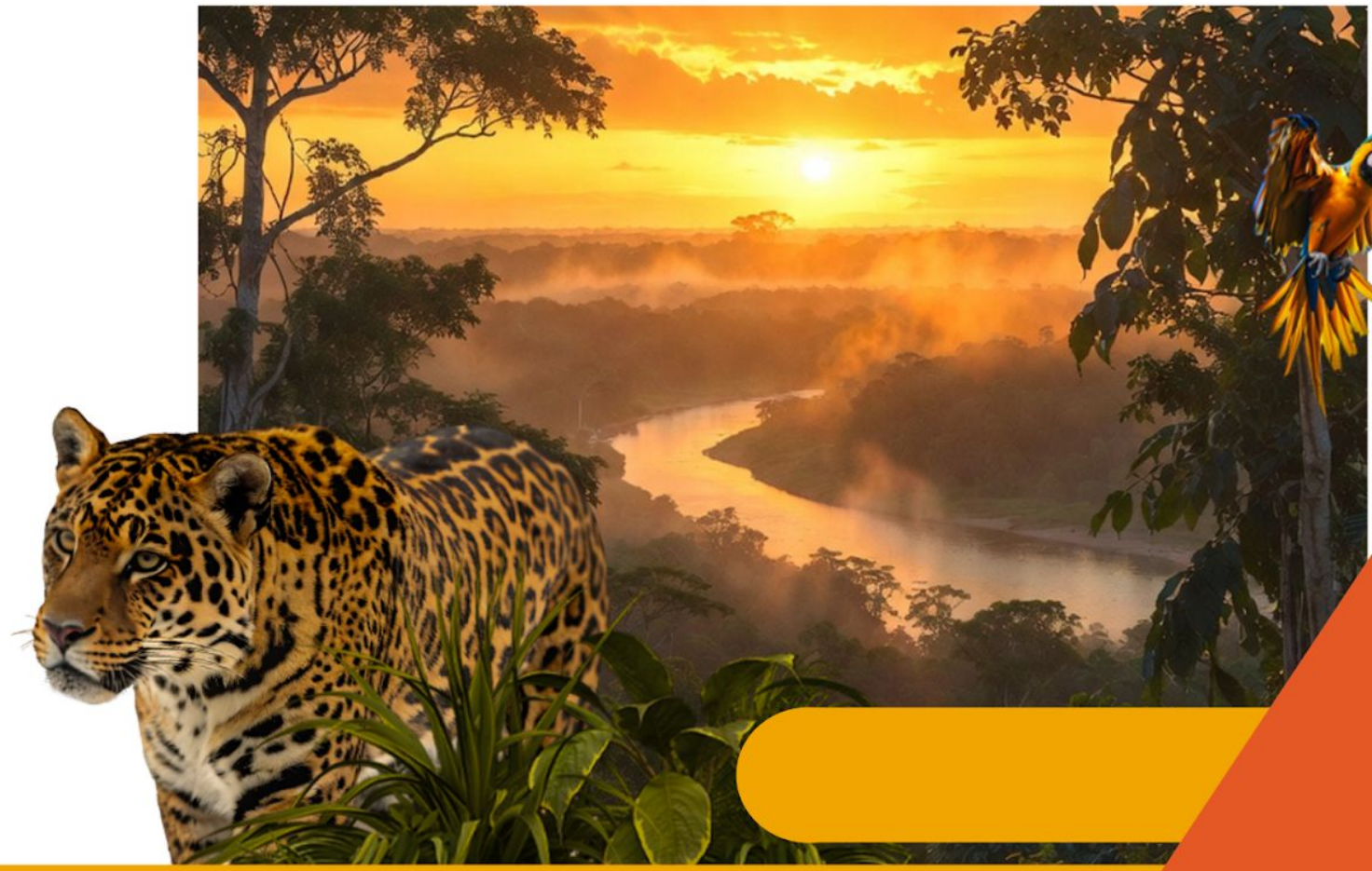
BDC Explorer

Gabriel Sansigolo

Brazilian National Institute for Space Research (INPE)



Overview

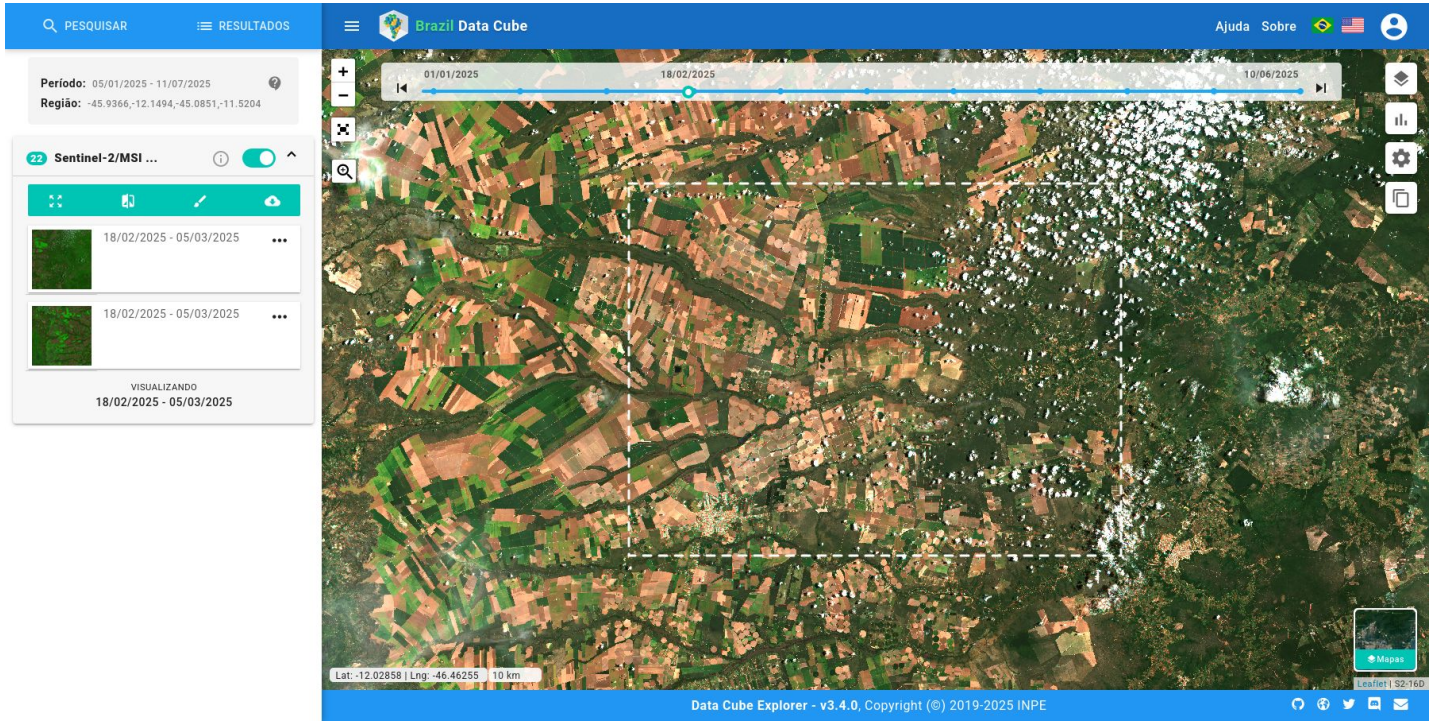
Brazil Data Cube Explorer is a web portal for **visualizing** data cubes, image collections and classifications.



https://data.inpe.br/bdc/explorer/

BDC Explorer

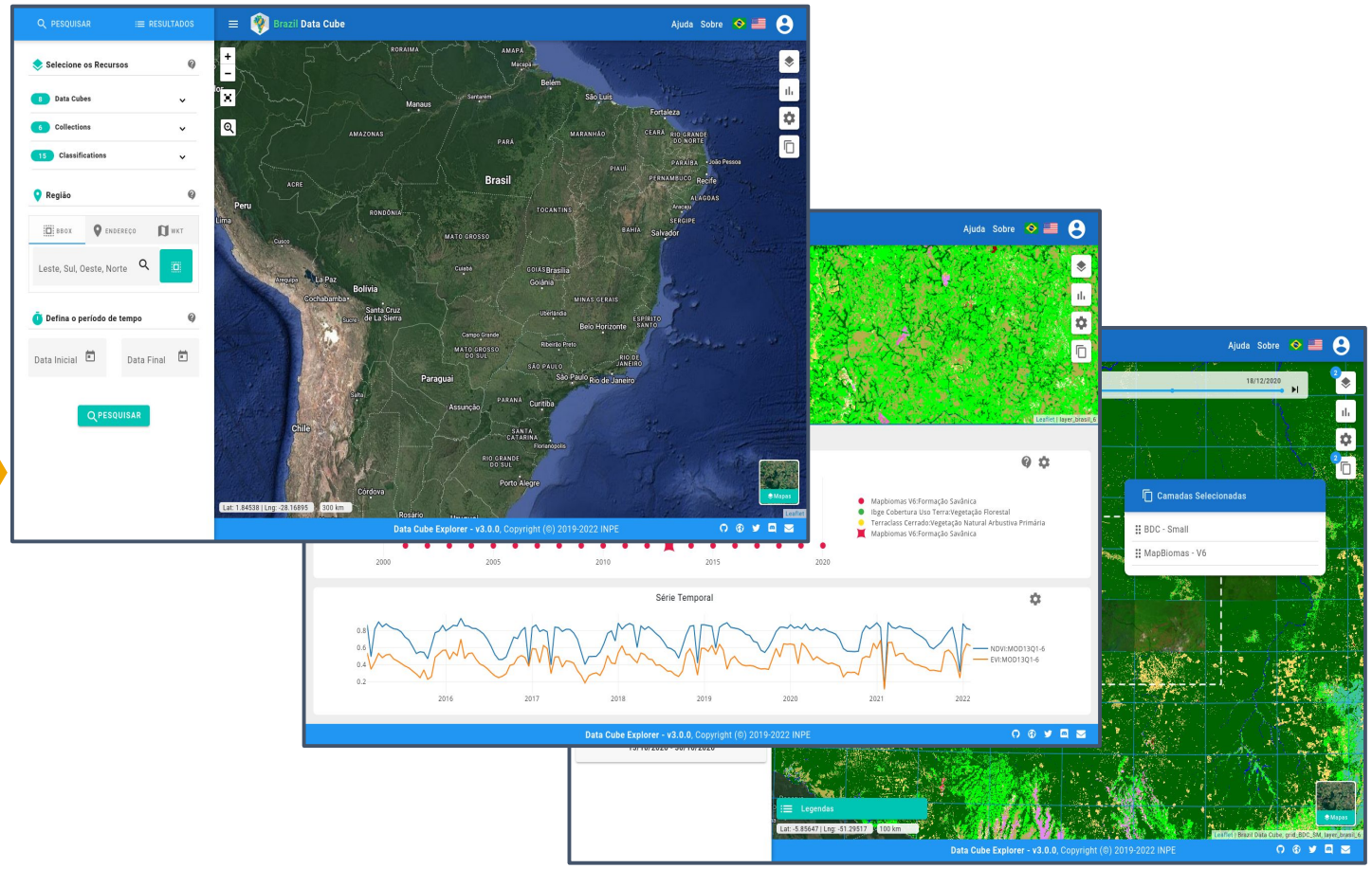
- BDC Explorer has a set of features for **viewing** and **downloading** image collections, some of which are:
  - Spatio-temporal search of data cubes, image collections and classifications;
  - Temporal visualization of images, with timeline;
  - Visualization of time series, land's trajectory and phenological metrics;
  - Editing the RGB composition of images;



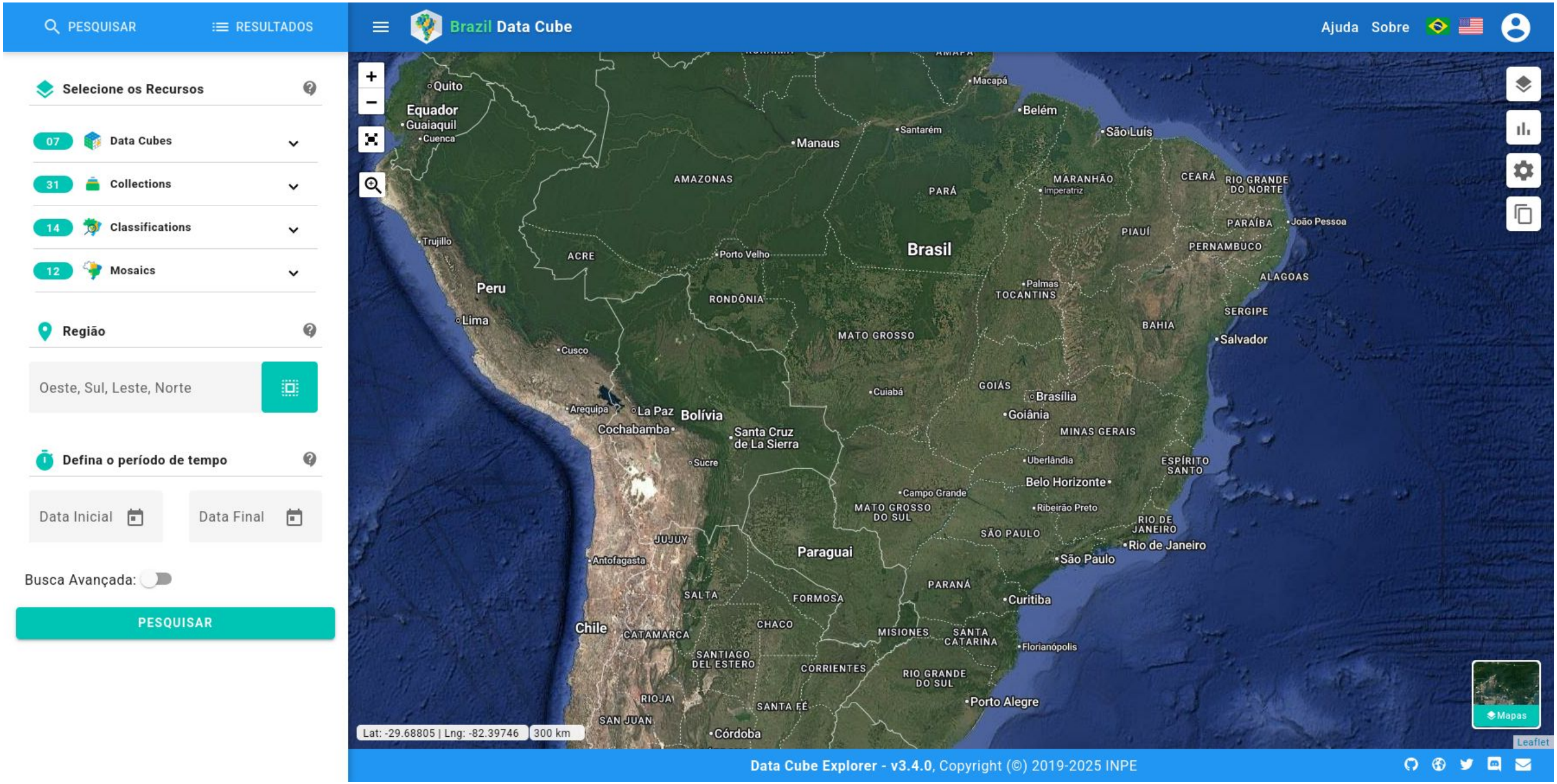
https://data.inpe.br/bdc/explorer/

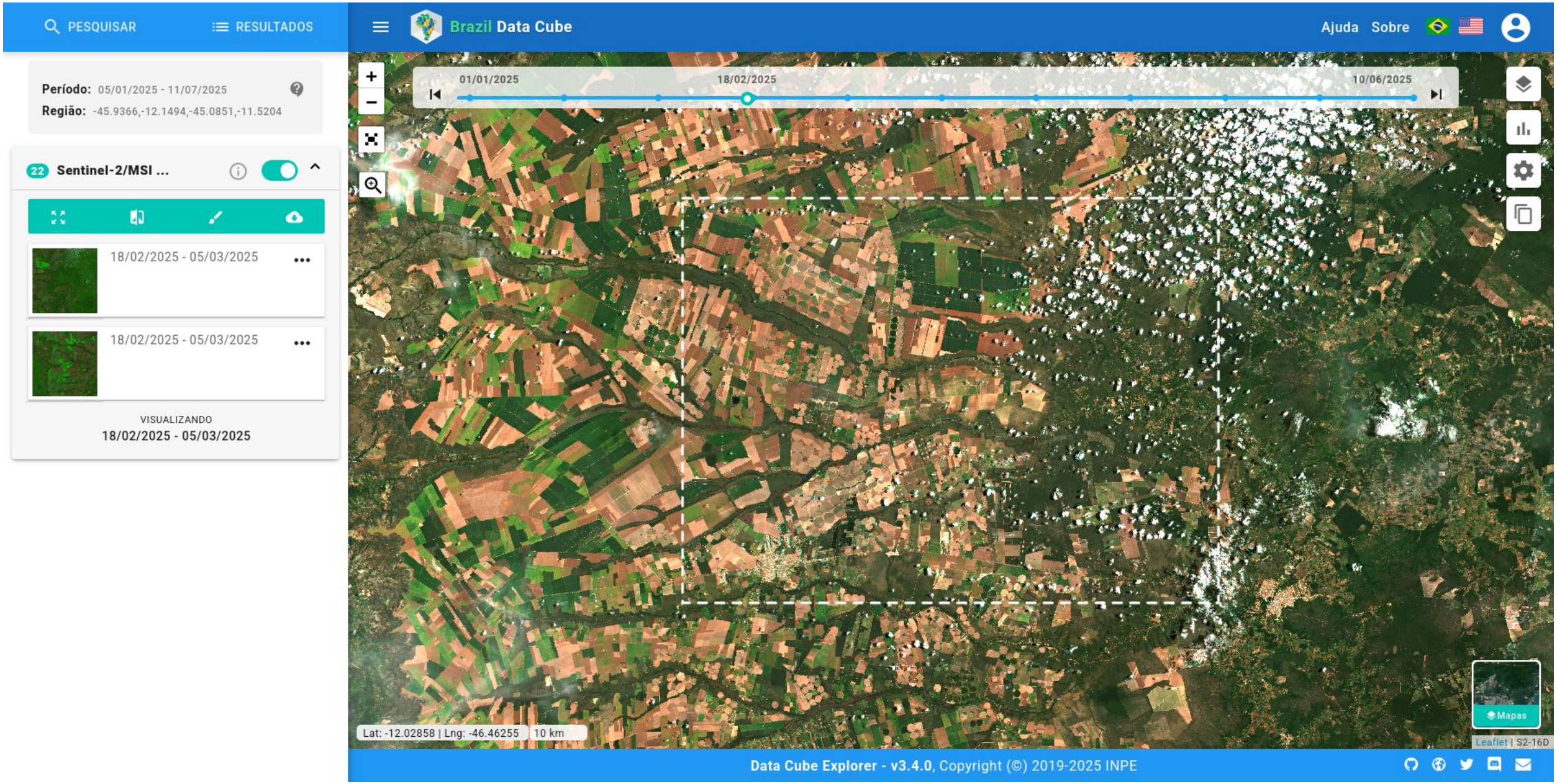
Summary

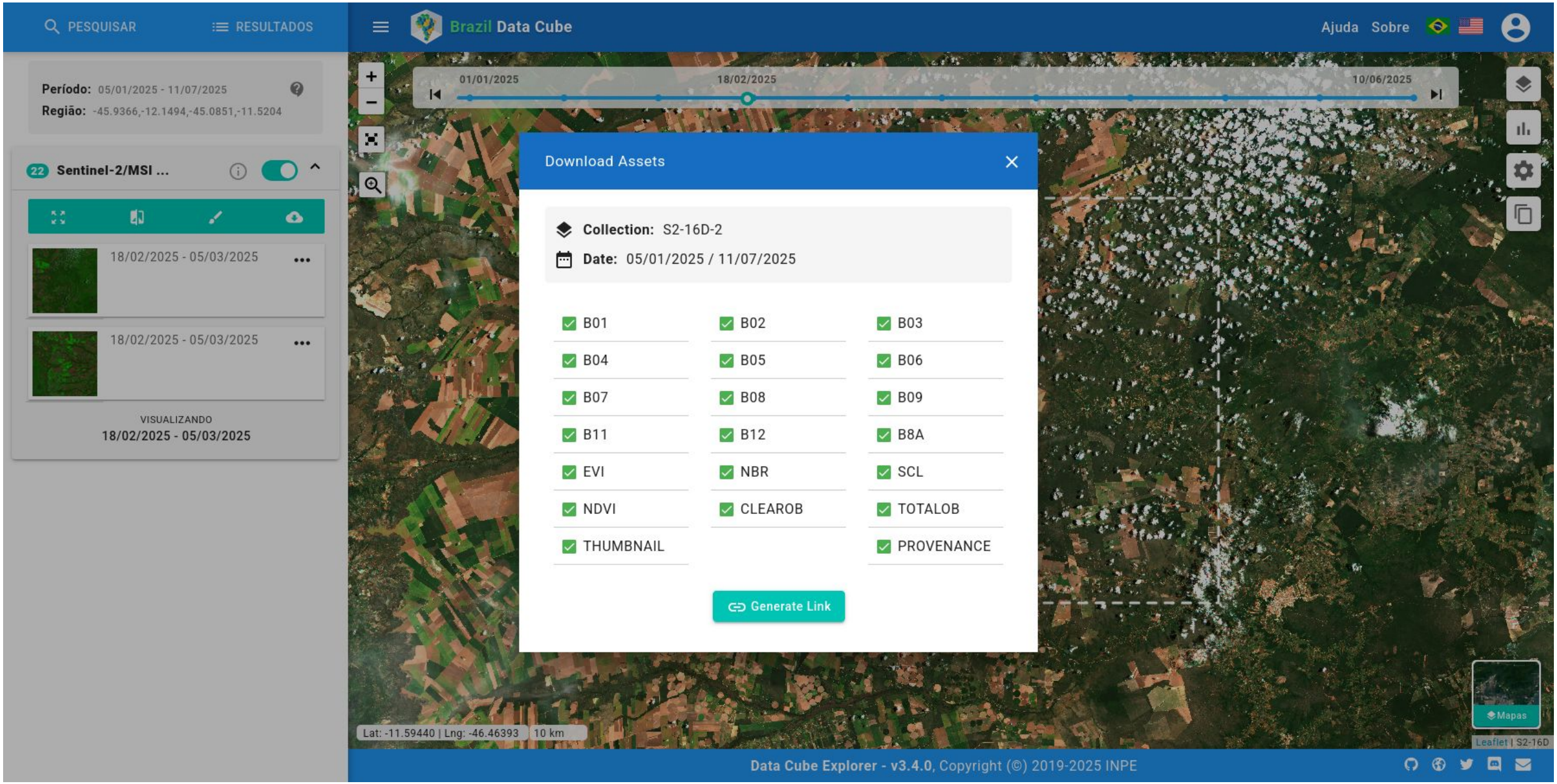
- For this part of the course, we will follow this outline:
  1. Search
  2. Results
  3. Style editing
  4. Download
  5. Time series
  6. Earth's trajectory
  7. Layers
  8. Visualization control
  9. Side-by-side visualization



https://data.inpe.br/bdc/explorer/


PESQUISAR
RESULTADOS
Brazil Data Cube
Ajuda
Sobre
Selecione os Recursos
07 Data Cubes
31 Collections
14 Classifications
12 Mosaics
Região
Oeste, Sul, Leste, Norte
Defina o período de tempo
Data Inicial
Data Final
Busca Avançada:
PESQUISAR
Quito
Equador
Guaiaquil
Cuenca
Macapá
Belém
Santarém
São Luís
Manaus
AMAZONAS
PARÁ
MARANHÃO
Imperatriz
CEARÁ
RIO GRANDE DO NORTE
PARAÍBA
João Pessoa
PIAUÍ
PERNAMBUCO
ALAGOAS
Trujillo
ACRE
Porto Velho
Brasil
Peru
RONDÔNIA
Palmas
TOCANTINS
SERGIPE
Lima
BAHIA
Salvador
MATO GROSSO
Cusco
Cuiabá
GOIÁS
Brasília
Goiânia
Arequipa
La Paz
Bolívia
Cochabamba
Santa Cruz de La Sierra
MINAS GERAIS
Sucre
Uberlândia
ESPÍRITO SANTO
Belo Horizonte
Campo Grande
MATO GROSSO DO SUL
Ribeirão Preto
RIO DE JANEIRO
SÃO PAULO
Rio de Janeiro
JUJUY
Paraguai
São Paulo
Antofagasta
PARANÁ
SALTA
FORMOSA
Curitiba
Chile
CHACO
CATAMARCA
MISIONES
SANTA CATARINA
Florianópolis
SANTIAGO DEL ESTERO
CORRIENTES
RIO GRANDE DO SUL
RIOJA
SANTA FÉ
Porto Alegre
SAN JUAN
Córdoba
Lat: -29.68805 | Lng: -82.39746
300 km
Mapas
Leaflet
Data Cube Explorer - v3.4.0, Copyright (©) 2019-2025 INPE


PESQUISAR
RESULTADOS
Período: 05/01/2025 - 11/07/2025
Região: -45.9366,-12.1494,-45.0851,-11.5204
22 Sentinel-2/MSI ...
18/02/2025 - 05/03/2025
18/02/2025 - 05/03/2025
VISUALIZANDO
18/02/2025 - 05/03/2025
Brazil Data Cube
Ajuda
Sobre
01/01/2025
18/02/2025
10/06/2025
Mapas
Lat: -12.02858 | Lng: -46.46255
10 km
Leaflet | S2-16D
Data Cube Explorer - v3.4.0, Copyright (©) 2019-2025 INPE


PESQUISAR
RESULTADOS
Brazil Data Cube
Ajuda
Sobre
Período: 05/01/2025 - 11/07/2025
Região: -45.9366,-12.1494,-45.0851,-11.5204
22 Sentinel-2/MSI ...
18/02/2025 - 05/03/2025
18/02/2025 - 05/03/2025
VISUALIZANDO
18/02/2025 - 05/03/2025
01/01/2025
18/02/2025
10/06/2025
Download Assets
Collection: S2-16D-2
Date: 05/01/2025 / 11/07/2025
B01
B02
B03
B04
B05
B06
B07
B08
B09
B11
B12
B8A
EVI
NBR
SCL
NDVI
CLEAROB
TOTALOB
THUMBNAIL
PROVENANCE
Generate Link
Lat: -11.59440 | Lng: -46.46393
10 km
Mapas
Leaflet | S2-16D
Data Cube Explorer - v3.4.0, Copyright (©) 2019-2025 INPE

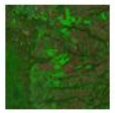

PESQUISAR
RESULTADOS
Período: 05/01/2025 - 11/07/2025
Região: -45.9366,-12.1494,-45.0851,-11.5204
22 Sentinel-2/MSI ...
18/02/2025 - 05/03/2025
18/02/2025 - 05/03/2025
VISUALIZANDO
18/02/2025 - 05/03/2025
Brazil Data Cube
Ajuda
Sobre
01/01/2025
18/02/2025
10/06/2025
Lat: -11.75284 | Lng: -45.75059
500 m
Leaflet | S2-16D
Ponto: -11.746033548688985, -45.743980407714844
Land Trajectory
2000
2005
2010
2015
2020
Terraclass Amazonia:No-Data
Mapbiomas V9:Formação Savânica
Mapbiomas V9:Mosaico de Usos
Mapbiomas V9:Soja
Ibge Cobertura Uso Terra:Vegetação Campest
Ibge Cobertura Uso Terra:Área Agrícola
Prodes Mata Atlantica:desmatamento
Time Series
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
2019
2020
2021
2022
2023
2024
2025
EVI:S2-16D-2
NDVI:S2-16D
Data Cube Explorer - v3.4.0, Copyright (©) 2019-2025 INPE

PESQUISAR RESULTADOS

Período: 05/01/2025 - 11/07/2025
Região: -45.9366,-12.1494,-45.0851,-11.5204

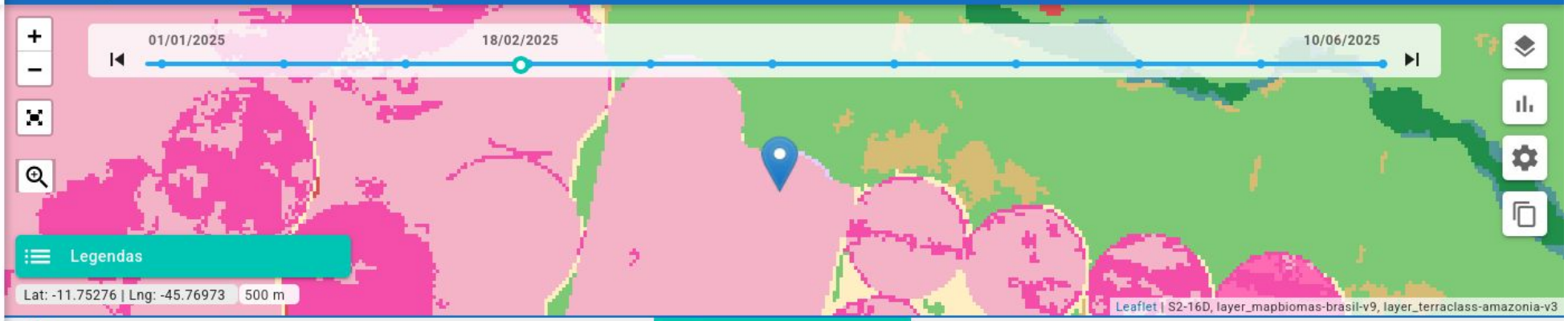
22 **Sentinel-2/MSI ...**


18/02/2025 - 05/03/2025


18/02/2025 - 05/03/2025

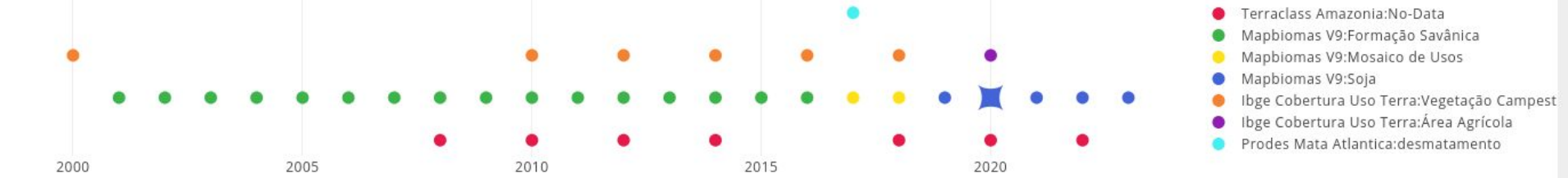
VISUALIZANDO
18/02/2025 - 05/03/2025

Brazil Data Cube — Ajuda Sobre

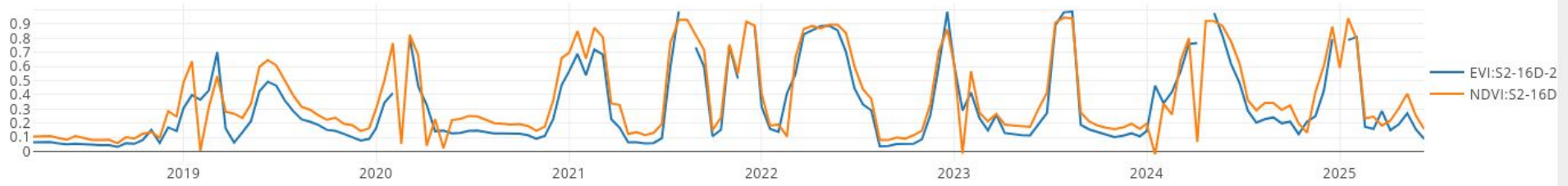


Ponto: -11.746033548688985, -45.743980407714844

Land Trajectory

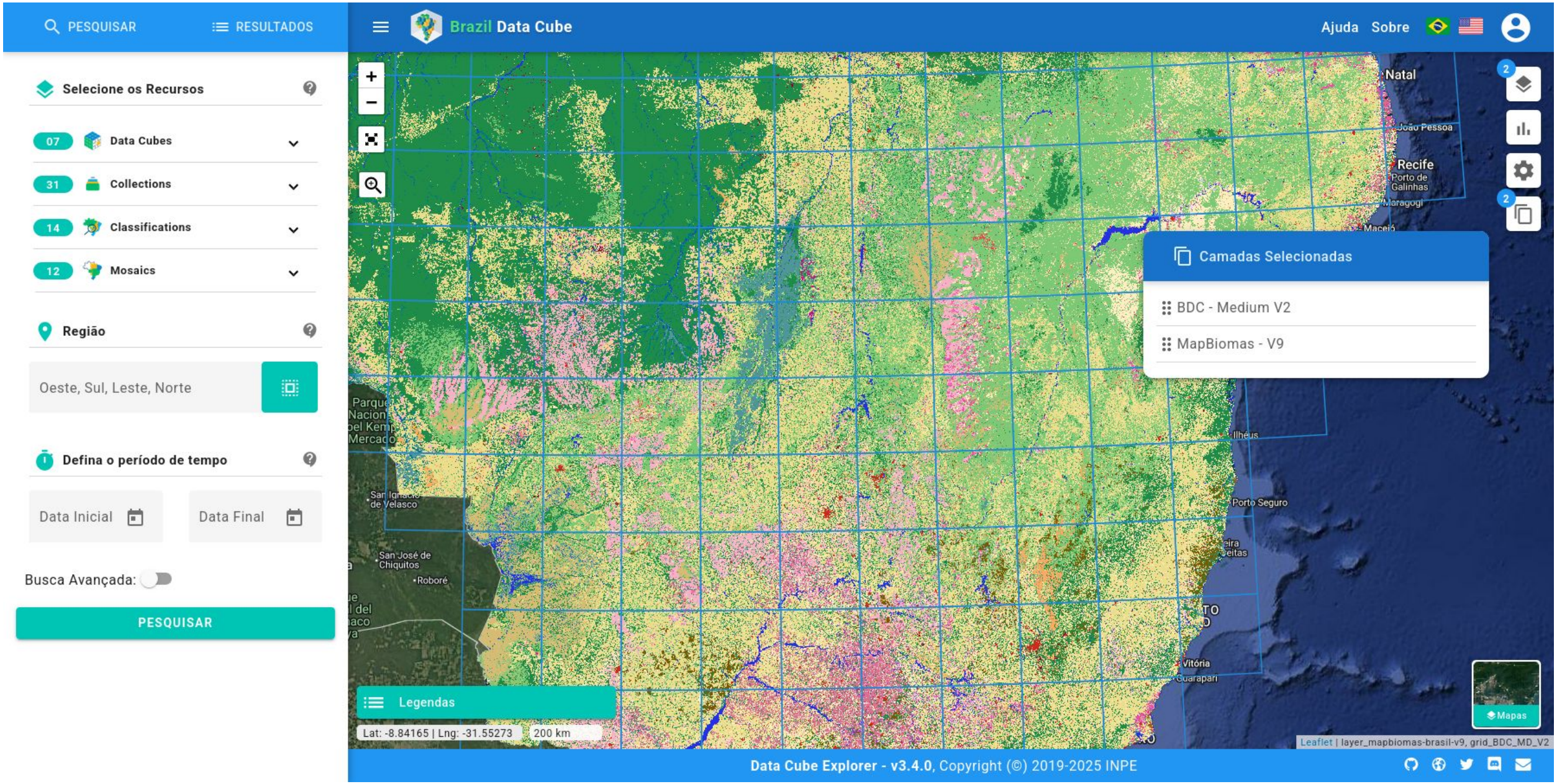


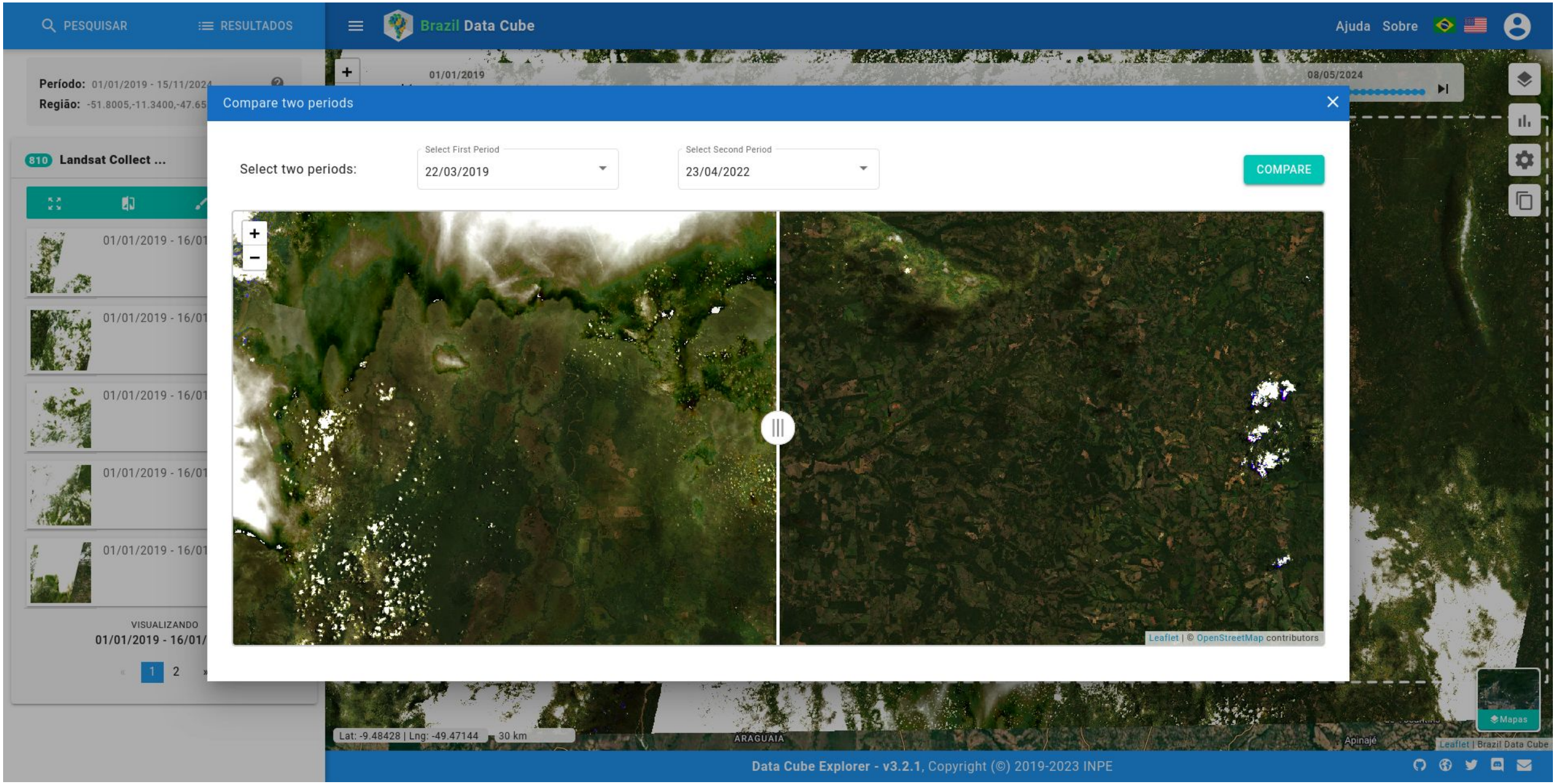
Time Series



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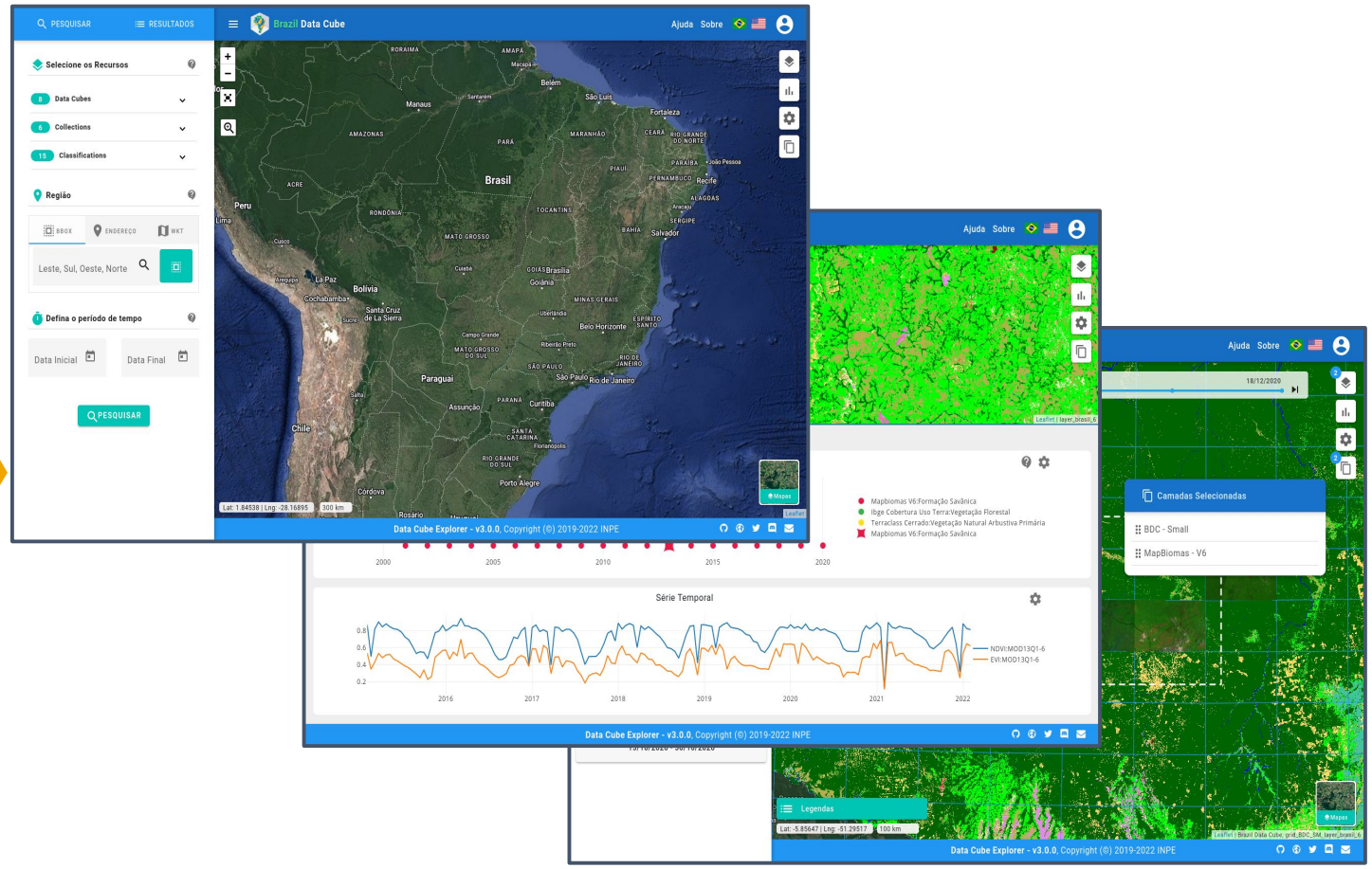

PESQUISAR
RESULTADOS
Período: 05/01/2025 - 11/07/2025
Região: -45.9366,-12.1494,-45.0851,-11.5204
22 Sentinel-2/MSI ...
18/02/2025 - 05/03/2025
18/02/2025 - 05/03/2025
VISUALIZANDO
18/02/2025 - 05/03/2025
Brazil Data Cube
Ajuda
Sobre
01/01/2025
18/02/2025
10/06/2025
Lat: -11.83274 | Lng: -45.76080
3 km
Leaflet | S2-16D
Ponto: -11.746033548688985, -45.743980407714844
Phenological Metrics
NDVI
Smooth ND
SOS
POS
EOS
VOS
Mar 2021
May 2021
Jul 2021
Sep 2021
Nov 2021
Data Cube Explorer - v3.4.0, Copyright (©) 2019-2025 INPE


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Busca Avançada:
PESQUISAR
Camadas Selecionadas
BDC - Medium V2
MapBiomas - V9
Natal
João Pessoa
Recife
Porto de Galinhas
Maceió
Ilhéus
Porto Seguro
Vitória
Guarapari
San José de Chiquitos
Roboré
Legendas
Lat: -8.84165 | Lng: -31.55273
200 km
Mapas
Leaflet | layer_mapbiomas-brasil-v9, grid_BDC_MD_V2
Data Cube Explorer - v3.4.0, Copyright (©) 2019-2025 INPE


PESQUISAR
RESULTADOS
Brazil Data Cube
Ajuda
Sobre
Período: 01/01/2019 - 15/11/2024
Região: -51.8005,-11.3400,-47.65
810 Landsat Collect ...
01/01/2019 - 16/01
VISUALIZANDO
01/01/2019 - 16/01/
01/01/2019
08/05/2024
Compare two periods
Select two periods:
Select First Period
22/03/2019
Select Second Period
23/04/2022
COMPARE
Leaflet | © OpenStreetMap contributors
Lat: -9.48428 | Lng: -49.47144
30 km
ARAGUAIA
Apinajé
Mapas
Leaflet | Brazil Data Cube
Data Cube Explorer - v3.2.1, Copyright (©) 2019-2023 INPE

Summary

- For this part of the course, we will follow this outline:
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  6. Earth's trajectory
  7. Layers
  8. Visualization control
  9. Side-by-side visualization



https://data.inpe.br/bdc/explorer/

Thank you!

gilberto.queiroz@inpe.br